IE 675b: Machine Learning

Assignment 1 (HWS 2026)

Prof. Dr. Rainer Gemulla
University of Mannheim
Chair of Practical Computer Science I

Task	Maximum	Points
1	4	.
2	4	.
3	6	.
5	4	.
6	3	.
Total	21	.

Rules for Submission

You must carefully follow the rules stated below for this assignment. An assignment that violates any of these rules will be graded as FAIL.

a) Submission

(i) Hand-in your solution via ILIAS until the date specified there. This is a hard deadline.

(ii) We strongly recommend to work on this assignment in pairs—i.e, with one (and only one) additional student—and then hand in a pair submission. Both students must submit exactly the same contents.

(iii) Solution files (such as notebooks and the solution PDF) must clearly state the name and matriculation number of both students right at the beginning.

b) Code

(i) You must submit all relevant code files (e.g., Python scripts) that we require to run your code and to replicate your results. In particular, you must submit all Python scripts that we provided to you.

(ii) You must include a PDF of every Jupyter notebook in which the most relevant cells are executed (i.e., make sure to disable excessively verbose cells that inflate the page count).

(iii) It suffices to fill out the "holes" that are marked in the code fragments.

c) Solution PDF

(i) You must provide your answers by filling out the provided solution LaTeX file. You may add any LaTeX packages that you absolutely need.

(ii) You must submit a single PDF file (do not submit LaTeX source code). All answers must be provided *inside* the boxes (i.e., no resizing of boxes, no overflow, no additional pages, no modification of font (size), ...) specified therein.

(iii) Name your solution PDF as follows:

```
a0[assignment number]-[ilias name 1]-[ilias name 2].pdf
```

For instance: `a01-aturing-jvneumann.pdf`

d) AI usage

(i) You must fill out and sign the AI declaration at the end of the solution file following the provided instructions.

(ii) You take responsibility for all solutions that you hand in. You must know and understand all solutions and be able to explain and defend them.

(iii) Excessive AI usage (such as handing in pure AI solutions) is considered cheating and will be reported to the examination board. Potential measures include failing the assignment, failing the course, or ex-matriculation.

Task 1: Training — 4 Points

We will use a preprocessed variant of the MNIST digits dataset in this assignment. The task is to classify hand-written digits. There is one class for each digit (i.e., classes $0, 1, 2, ..., 9$). The features represent a scanned image (28×28 pixels, values in $\{0, 1, \ldots, 255\}$). The dataset contains both training data ($\approx$ 6000 images per class) and test data ($\approx$ 1000 images per class). See `http://yann.lecun.com/exdb/mnist/` for more information.

a) Implement a function `nb_train` that trains a Naive Bayes classifier for categorical data using a symmetric Dirichlet prior and MAP parameter estimates.

A stub, description of the parameters and expected result can be found in the provided Python files. For example, you may assume that all features take values in $\{0, 1, \ldots, K-1\}$, where K is the number of possible values.

b) First, try out your function with a uniform Dirichlet prior ($\alpha = 1$) on the small example provided by us. Then try it with add-one smoothing ($\alpha = 2$).

c) If both results are correct, train your model on the MNIST digits dataset. You may need to work with a sample of the training data (the source code to do this is provided).

Note. Can you get your implementation to train the model on the full dataset in a reasonable time of, say, a few seconds? If not, don't worry and continue with a sample of the data.

Task 2: Prediction — 4 Points

a) Implement a function `nb_predict` that takes your model and a set of examples, and outputs the most likely label for each example as well as the log probability (confidence) of that label. As before, a stub, description of the parameters and expected results can be found in the provided Python files.

b) Again, start with our small example, check that your implementation is correct, then try it on the digits dataset.

Task 3: Experiments on MNIST digits data — 6 Points

a) Train your model with $\alpha = 2$ on the MNIST training dataset, then predict the labels of the MNIST test data using your model. What is the accuracy ($= 1 -$ misclassification rate) of your model?

b) Plot some test digits for each predicted class label (code provided). Can you spot errors? Then plot some misclassified test digits for each predicted class label (code provided). Finally, compute the confusion matrix (`https://en.wikipedia.org/wiki/Confusion_matrix`, code provided). Discuss the errors the model makes.

Note. In your report, document and discuss your experiments and findings.

Task 4: Model selection (optional) — 0 Points

a) Use cross-validation to find a suitable value of the hyperparameter α (of the symmetric Dirichlet prior).

b) Also plot the accuracy (as estimated via cross-validation) as a function of α. Discuss.

Note. Some hints to get you started with running cross-validation are given in the Python file.

Task 5: Generating data — 4 Points

a) Implement a function `nb_generate` that generates digits for a given class label. A stub, description of the parameters, and expected results can be found in the provided Python files.

b) Generate some digits of each class for your trained model and plot (code provided). Interpret the result. Repeat data generation for different models by varying the hyperparameter α. How does α influence the results? Discuss.

Task 6: Missing Data — 3 Points

Consider a classification problem with D discrete features, each taking values in $\{0, \dots, K-1\}$, and C classes $0, \dots, C-1$. For the MNIST dataset, we had $D = 784$, $K = 256$, and $C = 10$.

We make the Naive Bayes assumption and are given distributions $p(y)$ and $p(x_j|y)$ for $1 \leq j \leq D$. Using solely these distributions, give an **as simple as possible** formula for each of the following distributions and state why/when it can be useful

a) $p(y|x_{1:D})$

b) $p(y|x_{1:D'})$ for $1 \leq D' < D$.

c) $p(x_{D'+1:D}|x_{1:D'})$ for $1 \leq D' < D$.

Optional: Use the Naive Bayes model that you created in the previous tasks to sample from the above distributions. Experiment with some example images and discuss the result.